

# Vin Bhaskara

Senior Applied Scientist – LLMs and Foundation Models

IIT Silver Medalist | Vector AI Scholar | Prev: Samsung AI, University of Toronto

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Citizenship: Canadian, Location: Montréal

## SUMMARY

**Product-driven AI research, scalable engineering, measurable impact.** \$200MM+ production impact across 13 million customers at RBC, on-device denoising and super-resolution for Samsung Galaxy flagships. 11 papers, 3 patents, 300+ citations.

## WORK EXPERIENCE

**8+ years** of full-time experience in AI and software engineering

**Senior Applied Scientist**, Foundation Models and LLMs

**Borealis AI (RBC Research Institute), Montréal**

Deep Learning for Capital Markets and Credit Modeling at Canada's Largest Bank

Aug 2022 – **Present**

- Led R&D on fine-tuned masked language models and agentic RAG over Bloomberg chat across multiple trading desks, delivering **\$5MM+ per desk** annually and co-inventing a **diverse-retrieval RAG** method (US Patent Application 19/575,161)
- Co-led R&D for the flagship credit capacity product using **foundation models** trained on transaction data across **13 million customers**, driving ~\$200MM in 5-year NIBT with improved fairness and interpretability (US Patent Application 19/091,539)
- Shipped two additional **explainable** neural credit risk scoring models adding ~\$20MM projected annual NIBT – recognized with RBC Quarterly Team Performance Award (2026)

**Research Engineer**, Computer Vision

**Samsung AI Centre, Toronto**

Deep Learning for Image Enhancement and Synthesis

Feb 2020 – Jun 2022

- Co-invented **unsupervised super-resolution** training-data construction (US Patent 12,210,587) and developed **on-device digital-zoom** super-resolution for Samsung Galaxy flagship cameras (ECCV 2020 workshop paper)
- Led research on GraN-GAN (WACV 2022 paper): a piecewise gradient normalization method for **stable GAN training**, applied to image synthesis tasks inside Samsung's imaging pipeline
- Led research on multi-frame alignment for burst photography using neural implicit representations and **self-supervised blind image denoising** for low-light **night mode** – recognized with “Samsung Research America Rockstar” performance award (2021)

**Software Engineer 2**

**Broadcom Inc., Pune**

Machine Learning and Big Data for Malware Detection

Jul 2016 – Jul 2018

- Co-led a production **XGBoost model on Norton Anti-Virus** using Symantec's **big data file telemetry** that reduced previously missed malware detections by **> 60%** – recognized with Level 1 & 3 company-wide recognition awards
- Led research on image representations of dynamic run-time file behavior for GAN-based **proactive malware detection** and **software categorization** ([arXiv:stat.ML/1807.07525](https://arxiv.org/abs/1807.07525))

## EDUCATION

**M.Sc. in Applied Computing** (Deep Learning), Department of Computer Science, 4.0/4.0 (**A+**)

Sep 2018 – Dec 2019

**University of Toronto**, Downtown Toronto, Canada

- Received the **Vector Institute Scholarship in Artificial Intelligence (VSAI)** valued at \$17,500 awarded to **66 scholars in Ontario**
- Research: Part-Based Single-Shot Object Detection for Computer Vision. Supervisors: Dr. Alex Levinstein and Prof. Allan Jepson.

**B.Tech. in Engineering Physics** with Minor in **Electronics Engineering**, Department **Rank 1**

Jul 2012 – Jun 2016

**Indian Institute of Technology (IIT)**, Guwahati

- Institute Silver Medalist** for the best academic performance in the department among the graduating class of 2016 at IIT Guwahati
- Primary author of a [foundational paper](#) on Quantum Entanglement theory and a **visiting scholar** at the Institute for Quantum Computing (IQC), University of Waterloo, Canada

## RELEVANT PUBLICATIONS

Citations: **300+**, h-index: **8** - [Google Scholar](#)

- Vin Bhaskara**, H. Wang. “Curiosity-Critic: Cumulative Prediction Error Improvement as a Tractable Intrinsic Reward for World Model Training,” [arXiv:cs.LG/2604.18701](https://arxiv.org/abs/2604.18701) (Under Review) Apr 2026
- Vin Bhaskara\***, T.A. Armstrong\*, A. Jepson, A. Levinstein. “GraN-GAN: Piecewise Gradient Normalization for Generative Adversarial Networks,” [WACV 2022 Conference](#) (2022 IEEE Winter Conference on Applications in Computer Vision) Jan 2022
- Vin Bhaskara\***, H. Wang\*, A. Levinstein\*, S. Tsogkas, A. Jepson. “Efficient Super-Resolution Using MobileNetV3,” [ECCV 2020 Workshop](#) (2020 European Conference on Computer Vision Workshop) Jan 2021
- Vin Bhaskara**, S. Tsogkas, K. Derpanis, A. Levinstein. “Part-based Auxiliary Objectives with No Extra Labels for Robust Single-Shot Object Detection,” [dx.doi.org/10.13140/RG.2.2.10079.47521](https://dx.doi.org/10.13140/RG.2.2.10079.47521) (Preprint) Apr 2020

5. **Vin Bhaskara**, S. Desai. “Exploiting uncertainty of loss landscape for stochastic optimization,” [arXiv:cs.LG/1905.13200](https://arxiv.org/abs/cs.LG/1905.13200) (Preprint) May 2019
  6. **Vin Bhaskara**, Y. Fu, S. Gowda. “Risk Prediction in the General Internal Medicine Ward at St. Michael’s Hospital,” [dx.doi.org/10.13140/RG.2.2.27695.55205](https://doi.org/10.13140/RG.2.2.27695.55205) (Preprint) Apr 2019
  7. **Vin Bhaskara**, D. Bhattacharyya. “Emulating malware authors for proactive protection using GANs over a distributed image visualization of dynamic file behavior,” [arXiv:stat.ML/1807.07525](https://arxiv.org/abs/stat.ML/1807.07525) (Preprint) Jul 2018
- (\* Denotes equal contribution)

## GRANTED PATENTS

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1. H. Wang, X. Sun, **Vin Bhaskara**, S. Tsogkas, A. Jepson, A. Levinstein. “Unsupervised Super-Resolution Training Data Construction,” Samsung AI Centre Toronto, [US Patent 12,210,587](#) Jan 2025

## PATENT APPLICATIONS

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1. L. Schell, **Vin Bhaskara**. “Systems, Methods, and Techniques for Performing Retrieval Augmented Generation (RAG) with a Diverse-Retrieval Method,” Borealis AI (RBC Research), [US Patent Application No. 19/575,161 \(Filed Mar 23, 2026\)](#) Mar 2026
2. S.H. Hajimirsadeghi, M.O. Ahmed, E.J. Smith, **Vin Bhaskara**, *et al.* “Generating Credit Capacity Estimates,” Borealis AI (RBC Research), [US Patent Application No. 19/091,539 \(Filed Mar 17, 2025\)](#) Mar 2025

## ACADEMIC SERVICE

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- **Reviewer** for NeurIPS, CVPR, ICLR, and ICML 2023 – Present

## TECHNICAL SKILLS

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- **Languages:** Python (expert), Java, C++, C
- **AI Frameworks:** PyTorch, JAX, Hugging Face (Transformers, PEFT, Datasets), vLLM
- **Vector Databases:** FAISS, pgvector
- **Data and MLOps Tooling:** SQL, NoSQL, Hadoop, HDFS, MapReduce, Pandas, Docker, Weights & Biases

## AWARDS AND HONORS

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- **Vector Institute** Scholarship in Artificial Intelligence (VSAI) valued at \$17,500 – **1 of 66 scholars in Ontario** 2018
- **Institute Silver Medalist, IIT Guwahati** – best academic performance in Engineering Physics, class of 2016 2016
- Offered **AI Residency** at **Google X** (Mountain View) – declined 2019
- **Kaggle “Competitions Expert”** – Ranked Top 5% Global, 5 Medals 2017 – Present
- **Top 5%** (201st of 4,436 teams) in a solo submission, earning a **Kaggle Silver Medal** for predicting Nasdaq stock price movements on real market data 2024
- USEQIP Summer School at the Institute for Quantum Computing (IQC), Waterloo, Canada – **25 students selected globally** 2015

## OTHER PUBLICATIONS

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1. **Vin Bhaskara**\*, S.N. Swain\*, P.K. Panigrahi. “Generalized Entanglement Measure for Continuous-Variable Systems,” [Physical Review A \(PRA\) 105, 052441 \(2022\)](#), American Physical Society Journals May 2022
2. **Vin Bhaskara**, P.K. Panigrahi. “Generalized concurrence measure for faithful quantification of multiparticle pure state entanglement using Lagrange’s identity and wedge product,” [Quantum Inf. Process. 16 \(5\), 118](#), Springer Journals Mar 2017
3. J. Flannery, G. Bappi, **Vin Bhaskara**, O. Alshehri, M. Bajcsy. “Implementing Bragg mirrors in a hollow-core photonic crystal fiber,” [Optical Materials Express 7 \(4\), 1198](#), Optical Society of America Journals Mar 2017
4. C.M. Haapamaki, J. Flannery, G. Bappi, R. Al-Maruf, **Vin Bhaskara**, O. Alshehri, T. Yoon, M. Bajcsy. “Mesoscale cavities in hollow-core waveguides for quantum optics with atomic ensembles,” [Nanophotonics 5 \(1\)](#), De Gruyter Journals Sep 2016

(\* Denotes equal contribution)